Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18ME53]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2021

DESIGN OF MACHINE ELEMENTS II

Time: 3 Hours **Maximum Marks:70**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
- **ii.** Missing data, if any, may be suitably assumed.
- **iii.** Use of data hand books is permitted

1. a. Present the step wise procedure involved in design of a V Belt drive **07**

b. A flat leather belt is required to transmit 20 kW from a pulley of 1.2 m diameter running at 250 rpm to another pulley that runs at 500 rpm. On taking the coefficient of friction as 0.25, safe working stress for leather as 2 MPa, thickness of belt as 10 mm, determine the width of the belt. Assume specific weight of leather as 10×10^{-6} N/mm^3. **07**

2. a. Define the following terms of spur gear drive system **05**

- **i.** Circular pitch
- **ii.** Diametral pitch
- **iii.** Module

b. A spur pinion made of SAE 1040 steel and rotating at 400 rpm drives a high grade cast iron gear. The transmission ratio is 4:1. The pinion has 15 standard, 20°. full depth involute teeth of 4 mm module. Face width of both the wheels is 40 mm. How much power can be transmitted by the gear system from the standpoint of strength. Take bending strength of SAE 1040 steel and high grade cast iron as 173 MPa and 103 MPa respectively. **09**

3. a. Sequentially list the steps involved in obtaining the module of bevel gear tooth. **07**

b. Find the module of pair of bevel gears used to connect two shafts at 60 deg. The gears are made of alloy steel of case hardened (SAE2320) and precision cut with form cutter. The gear ratio is 5:1. The power transmitted is 30 kW at 900 rpm of pinion. The teeth are 20°. full depth involute. The pinion has 24 teeth. Assume service factor of 1.5 and bending strength of SAE2320 steel as 345 MPa. **07**

OR

4. a. Knowing the module, sequentially list the steps involved in obtaining the hardness of bevel gear tooth **07**

	b.	Obtain the module of bevel gears that transmit power of 25 kW from a shaft rotating at 1200 rpm to a perpendicular shaft to be rotated at 400 rpm. Take bending strength of pinion and gear materials as 345 MPa and 220 MPa respectively. Assume pressure angle of 20° full depth involute and service factor as 1.5.	07
5.	a.	Define a spring and state its applications.	05
	b.	Design a helical compression spring to support an axial load of 3000 N. The deflection under load is limited to 60 mm. Spring index is 6. The spring is made of Chrome Vanadium steel whose shear strength is 690 MPa. Assume modulus of rigidity of steel as 79340 N/mm^2 and factor of safety of 2.	09

OR

6.	a.	State the advantages of helical springs. Name various types of springs. .	07
	b.	Design the helical spring used in a recoil system so as to absorb 120 Nm of energy under maximum force of 3000 N. Take shear strength of spring wire as 550 MPa and factor of safety of 2. Consider spring index as 8 and modulus of rigidity of spring steel as 79340 N/mm^2.	07
7.	a	Define hydrodynamic lubrication and list the requirements of a good lubricant.	06
	b	Determine the power loss for a Petroff bearing 100 mm in diameter and 150 mm long. The radial clearance is 0.05 mm. Speed of the journal is 1000 rpm. The lubricating oil is SAE 10 and bearing operating temperature is 60 °C.	08

OR

8.	a.	Define rolling contact bearing and name various types of such bearings.	06
	b.	A taper roller bearing has a dynamic load capacity of 26 kN. The desired life for 90% of the bearings is 8000 h and the speed is 300 rpm. Calculate the equivalent radial load that the bearing can carry.	08